Separating the Bound and Mean Steps of Post-Training Repair for Reaction–Diffusion PINNs

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Abstract

Physics-informed neural networks are widely used for reaction–diffusion problems, but exported fields may violate bounds or a prescribed spatial mean. Post-training mean–box projection restores feasibility, yet prior work typically reports one combined before/after comparison and does not separate clipping from mean adjustment. On periodic stiff heat–species reaction–diffusion we run a held-out paired experiment with ten seeds and three evaluation modes: raw output, box clipping alone, and full 60-step ET-MBP. Box clipping removes held-out upper-bound violation from 11.2% to zero without measurable change in Rel.$L_2(c)$, PDE p_{95}, or the coupled-mean invariant; measurable costs are confined to the mean step, with Rel.$L_2(c)$, PDE p_{95}, and invariant error increasing by $(3.05 \pm 1.18) \times 10^{-3}$, 0.146 ± 0.052, and 0.018 ± 0.002, respectively. Converged capped-simplex projection reproduces the 60-step values, so the cost reflects the closure target rather than iteration error.

Index Terms

physics-informed neural networks, post-training repair, alternating projection, reaction–diffusion

I. Introduction

Physics-informed neural networks provide continuous surrogates for partial differential equations [1], [2], but stiffness and sharp fronts can produce unstable optimization and inadmissible fields [3]–[6]. Classical IMEX and positivity-preserving discretizations control such behavior when the numerical operator is available [7], [8]. A trained neural surrogate, however, may already have been exported when a bound violation is detected.

Constraints can be imposed during training through conservation or structure-preserving losses [9]–[11], through hard network maps [12], differentiable optimization [13] and KKT projection layers [14], [15], or after training by projecting the exported output [16], [17]. When a surrogate is already trained, post-training repair applies directly to the exported field. Published projection studies report whether a repaired field meets the target constraints and how accuracy changes relative to the raw network; they typically present one combined projection rather than separating interval clipping from mean adjustment, even though the two operations depend on different inputs.

The box-plus-fixed-sum set used here is a capped simplex with standard exact projection algorithms [18]. The projection operator is adopted as an established feasibility restoration technique.

Separating the two steps matters because box clipping requires only the admissible interval, whereas mean adjustment requires a numerical closure value that, for a stiff reaction, is available only as a biased estimate. Reporting them together attributes any accuracy change to feasibility as a whole. Here an explicit box-only evaluation is inserted between the raw and fully repaired fields so that bound removal, mean enforcement, and their respective costs can be measured independently on held-out time slices.

The experiment below operates on already-saved networks. Training mean loss and evaluation-time treatment are independent factors; all evaluations share the same architecture, labels, collocation schedule, and random seeds. Accuracy is measured only on time slices absent from labeled training, while closure, PDE residual, heat-source error, and the exact coupled-mean invariant are reported separately.

This paper makes three contributions:

- It evaluates box clipping and mean adjustment as separate post-training operations rather than one composite repair.
- It reports a held-out paired design that separates training-side mean loss from evaluation-side repair on ten random seeds.
- It locates every measurable repair cost in the mean step while box clipping removes bound violations without a resolvable penalty in accuracy, PDE consistency, or the exact coupled-mean invariant on the benchmark studied here.

II. METHODS

A. Model problem

On $(x,t) \in [0,1] \times [0,0.5]$ with periodic boundary conditions,

$$\partial_t c = D_c \partial_{xx} c + k_r c(1-c), \tag{1}$$

$$\partial_t T = D_T \partial_{xx} T + \alpha k_r c(1-c), \tag{2}$$

where $D_c = 0.003$, $D_T = 0.025$, $k_r = 25$, and $\alpha = 1.2$. The reaction rate k_r makes the species equation stiff on the chosen horizon, while the coupling parameter α links the exothermic source in (2) to the autocatalytic species field in (1). Initial fields are

$$T(x,0) = 1 + 0.1\sin(2\pi x), \qquad c(x,0) = 0.45 + 0.2\cos(2\pi x).$$

The finite-difference reference has 500 periodic nodes and 250 saved times with CFL subcycling. Stored $2\times$ and $4\times$ refinement comparisons give maximum cross-grid differences 9.89×10^{-4} and 4.92×10^{-4}, respectively.

Let $M_c = \langle c \rangle$ and $M_T = \langle T \rangle$. Periodicity gives the exact identities

$$\dot{M}_c = k_r\{M_c(1-M_c) - \mathrm{Var}(c)\}, \tag{3}$$

$$\frac{d}{dt}(M_T - \alpha M_c) = 0. \tag{4}$$

Dropping the variance term yields the first-moment mean-field target

$$M_c^\star(t) = \frac{M_c(0)}{M_c(0) + [1 - M_c(0)]e^{-k_r t}}. \tag{5}$$

Because the omitted term in (3) is non-positive, this target overestimates conversion for heterogeneous fields. Its held-out discrepancy from the finite-difference mean, E_{model}, is 4.21×10^{-3}; Eq. (4), by contrast, is exact.

B. Evaluation metrics

For seed s and held-out time set $\mathcal{H}$, define

$$E_{\mathrm{closure}}^{(s)} = \max_{t \in \mathcal{H}} |M_c^{(s)}(t) - M_c^\star(t)|, \tag{6}$$

$$E_{\mathrm{true}}^{(s)} = \max_{t \in \mathcal{H}} |M_c^{(s)}(t) - M_{c,\mathrm{FD}}(t)|, \tag{7}$$

$$E_{\mathrm{model}} = \max_{t \in \mathcal{H}} |M_c^\star(t) - M_{c,\mathrm{FD}}(t)|. \tag{8}$$

Triangle and reverse-triangle inequalities give

$$\max(0, E_{\mathrm{model}} - E_{\mathrm{closure}}^{(s)}) \le E_{\mathrm{true}}^{(s)} \le E_{\mathrm{model}} + E_{\mathrm{closure}}^{(s)}. \tag{9}$$

Consequently, forcing closure consistency can remove a raw-field error that had partially cancelled target bias. The three quantities above are reported separately to keep algebraic feasibility distinct from predictive accuracy.

C. Post-training repair

For a spatial vector $\boldsymbol{c}$ at one time, write the box step and the mean step as

$$B(\boldsymbol{c}) = \Pi_{[0,1]^N}(\boldsymbol{c}), \qquad H(\boldsymbol{c}) = \boldsymbol{c} - \bar{c}\,\mathbf{1} + M_c^\star \mathbf{1}. \tag{10}$$

ET-MBP applies $\boldsymbol{c} \leftarrow B(H(\boldsymbol{c}))$ for $N_{\mathrm{iter}} = 60$. These are projections onto a box and onto a mean hyperplane; classical alternating-projection convergence applies when their intersection is nonempty [19]. A finite iterate is an approximate feasibility repair and is not guaranteed to equal the Euclidean projection onto the intersection, so a numerically converged dual-bisection capped-simplex projection also serves as a post-hoc diagnostic. When the two constraints are feasible, the exact capped-simplex projection gives the Euclidean projection onto their intersection; agreement between the 60-step iterate and the converged projection therefore indicates that any remaining error is due to the target rather than to stopping early.

The two steps differ in what they require. B depends only on the admissible interval fixed by the problem statement, whereas H depends on the numerical value $M_c^\star(t)$ and therefore inherits E_{model}. Applying B alone gives a third evaluation that uses no target. The full alternating operator is referred to as evaluation-time mean–box projection, ET-MBP.

TABLE I
HELD-OUT RESULTS OVER TEN PAIRED SEEDS (MEAN±STD).

Configuration	Rel.$L_2(c)$	Viol.	E_{closure}	E_{true}
No-mean / raw	$(6.78 \pm 1.62) \times 10^{-3}$	0.112 ± 0.109	0.019 ± 0.003	$(4.03 \pm 2.47) \times 10^{-3}$
No-mean / box only	$(6.74 \pm 1.65) \times 10^{-3}$	0	0.019 ± 0.003	$(4.03 \pm 2.47) \times 10^{-3}$
No-mean / ET-MBP	$(9.78 \pm 0.67) \times 10^{-3}$	0	$(2.43 \pm 0.77) \times 10^{-6}$	0.019 ± 0.000
Mean-trained / raw	$(7.12 \pm 2.11) \times 10^{-3}$	0.158 ± 0.204	0.016 ± 0.002	$(4.30 \pm 2.43) \times 10^{-3}$
Mean-trained / box only	$(6.72 \pm 1.43) \times 10^{-3}$	0	0.016 ± 0.002	$(4.28 \pm 2.40) \times 10^{-3}$
Mean-trained / ET-MBP	0.010 ± 0.001	0	$(2.80 \pm 1.45) \times 10^{-6}$	0.019 ± 0.000

III. EXPERIMENTS

A. Setup

Both trained models use the same strictly periodic Fourier-feature MLP with six frequencies, four hidden layers of width 128, and $\tanh$ activation, AdamW at 2×10^{-3}, 3500 epochs, 18 000 labeled points, and 6000 PDE collocation points. Space enters only through $\sin(2^{j+1}\pi x)$ and $\cos(2^{j+1}\pi x)$, while time still enters both directly and through its own Fourier features. With L_T, L_c denoting point losses and L_{snap} a 500-point spatial-snapshot loss sampled each epoch, the objective is

$$L = 0.55(L_T + 3L_c) + 0.45L_{\text{snap}} + 0.12L_{\text{PDE}} + 0.05L_{\text{ub}} + 0.05L_{\text{nn}} + \lambda_{\text{mc}}L_{\text{mc}},$$

where the only training factor is the species-mean weight $\lambda_{\text{mc}} \in \{0, 0.25\}$.

The evaluation forms a 2×3 factorial design. Training uses either no species-mean loss or a soft mean loss; each saved network is then evaluated without repair, after box clipping alone, or after the full 60-step ET-MBP repair. All other choices are fixed across seeds, so paired differences isolate the effect of each repair step rather than architecture or sampling noise.

Every fifth time slice beginning at index 4 is held out: 50 of 250 slices, or 20%, including $t = 0.5$. They are excluded from both labeled-point sampling and the full-spatial-snapshot term. PDE collocation still spans the continuous full horizon, so the network continues to see the full time domain through the physics term even though labeled data are missing on $\mathcal{H}$. This split tests whether repair effects persist on time slices that never entered the supervised loss, rather than on slices that were heavily sampled during training. For each of ten paired seeds, the two saved raw networks are then evaluated three times: unchanged, after the box step B alone, and after the full 60-step repair. No lower clipping is applied anywhere, and none is needed, since the held-out non-negativity violation is zero in all six configurations. Averaged over seeds, the largest periodic endpoint mismatch in the no-mean arm is 2.44×10^{-7} in value and 9.90×10^{-7} in first derivative; the mean-trained arm gives residuals of the same order.

Channel-wise Rel.$L_2(c)$ and Rel.$L_2(T)$, upper-bound violation fraction for $c > 1 + 10^{-3}$, E_{closure}, E_{true}, and heat-source error are evaluated on $\mathcal{H}$. Relative L_2 error measures field accuracy against the finite-difference reference on withheld times. The violation fraction reports how much of the held-out grid exceeds the upper bound by the chosen tolerance. E_{closure} and E_{true} quantify mean-target consistency and mean accuracy, respectively, while the heat-source error tests whether the repaired field preserves the physically integrated reaction rate rather than its closure approximation.

The heat quantity integrates the physical source $k_r\langle c(1-c)\rangle$, not the closure $k_r M_c(1-M_c)$. PDE p_{95} is the 95th percentile after concatenating the T and c equation residuals, using a periodic centered stencil pooled over the full grid rather than over $\mathcal{H}$ alone. For scale, the same stencil applied to the FD reference itself gives 2.23×10^{-3}, and the 95th percentile of the reaction term is 5.84×10^{0}, so the raw networks sit two orders of magnitude above the discretisation floor at 0.05 of the reaction scale. The invariant metric is the held-out maximum error in $M_T - \alpha M_c$ relative to FD.

Table I lists all six evaluation configurations. The two raw rows report the saved networks before any post-processing. The two box-only rows insert clipping between raw and full repair and show that bound violations disappear while E_{closure} remains at the raw value, confirming that box clipping does not enforce the mean target. The two ET-MBP rows then drive E_{closure} to near zero and make E_{true} equal to the target bias by construction.

B. Box clipping results

Table I places the box-only evaluation between the raw and fully repaired rows. Clipping alone removes the held-out upper-bound violation, from 11.2% to zero without mean-loss training and from 15.8% to zero with it. Its paired effects on held-out

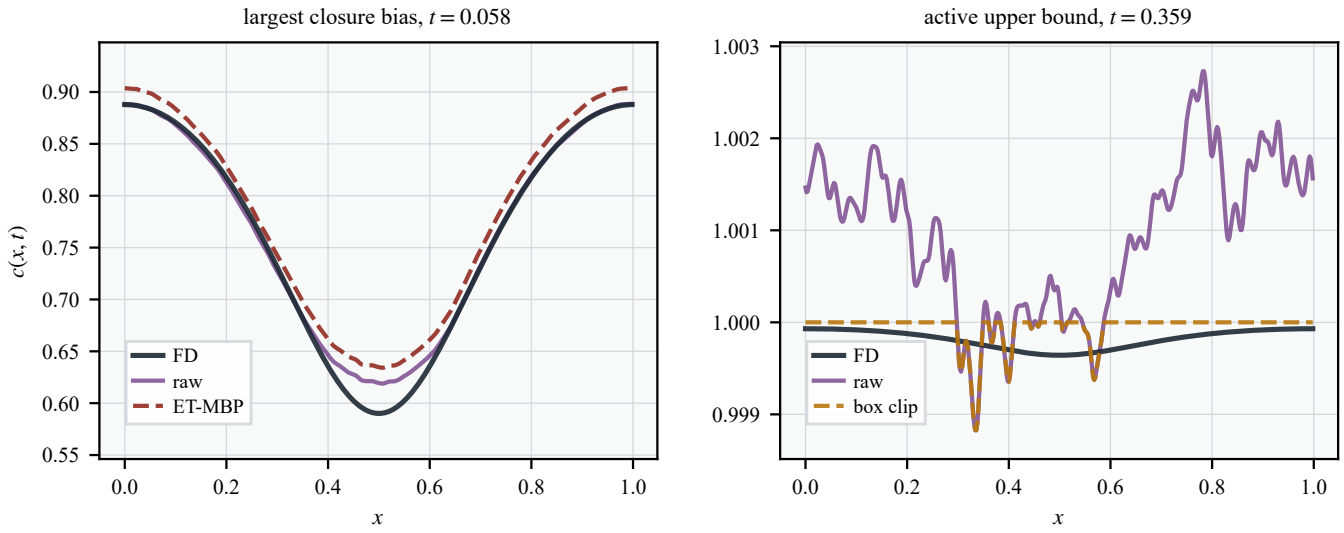


Fig. 1. Representative mean-trained seed on two held-out slices. Left: largest closure bias; ET-MBP shifts the profile away from the reference while the raw field remains close. Right: active upper bound; the raw overshoot sits above a saturated reference and box clipping returns it.

TABLE II
REPAIR COSTS OVER TEN PAIRED SEEDS (MEAN±STD).

Configuration	Heat error	PDE p_{95}	Invariant error
No-mean / raw	$5.7 \pm 6.4\%$	0.264 ± 0.080	$(4.44 \pm 3.23) \times 10^{-3}$
No-mean / box only	$5.2 \pm 6.7\%$	0.263 ± 0.080	$(4.49 \pm 3.19) \times 10^{-3}$
No-mean / ET-MBP	$5.8 \pm 0.2\%$	0.409 ± 0.013	0.023 ± 0.001
Mean-trained / raw	$5.2 \pm 5.9\%$	0.250 ± 0.090	$(4.25 \pm 2.81) \times 10^{-3}$
Mean-trained / box only	$3.3 \pm 2.4\%$	0.236 ± 0.053	$(4.19 \pm 2.62) \times 10^{-3}$
Mean-trained / ET-MBP	$5.7 \pm 0.3\%$	0.391 ± 0.019	0.021 ± 0.001

Rel.$L_2(c)$, E_{true}, PDE p_{95}, and the coupled invariant are $(-4.59 \pm 3.51) \times 10^{-5}$, $(1.40 \pm 3.17) \times 10^{-6}$, $(-0.73 \pm 1.30) \times 10^{-3}$, and $(0.54 \pm 1.20) \times 10^{-4}$ at mean difference $\pm 95\%$ CI, all of which contain zero. The mean-trained arm gives the same pattern.

The fields indicate why. Held-out slices on which the raw network exceeds the bound are late slices where the reference has already saturated at $c = 1$, so the excess is a small overshoot of a flat profile and clipping returns it to the reference; see the right panel of Fig. 1. On this benchmark the box step is active only where the reference itself lies at the bound.

C. Mean adjustment results

The costs in Table II are confined to the second step. Relative to the box-only field, the mean step raises held-out Rel.$L_2(c)$ by $(3.05 \pm 1.18) \times 10^{-3}$, E_{true} by 0.015 ± 0.002, pooled PDE p_{95} by 0.146 ± 0.052, and the error in the exact invariant (4) by 0.018 ± 0.002; with mean-loss training the four effects are $(3.36 \pm 0.64) \times 10^{-3}$, 0.015 ± 0.002, 0.155 ± 0.033, and 0.017 ± 0.002. Table II complements Table I by reporting the heat-source, PDE, and invariant metrics that are most sensitive to mean adjustment while leaving the temperature channel essentially unchanged across repair arms.

Two of them follow from Section II-B once the mean is forced: E_{true} becomes E_{model} identically, and shifting M_c while leaving M_T untouched perturbs $M_T - \alpha M_c$ by roughly α times the same bias. The PDE residual does not follow in that way, and it rises from 0.05 to 0.07 of the reaction scale. Heat-source error has large seed dispersion before repair, and its mean change is small relative to that dispersion, so no change can be claimed in either direction.

Figure 2 shows the two sets of effects side by side. The interaction contrasts between mean-loss training and the full repair, $(-2.93 \pm 5.13) \times 10^{-4}$ for Rel.$L_2(c)$ and $(-0.48 \pm 2.97) \times 10^{-7}$ for E_{true}, both include zero, so at this sample size the soft training term does not measurably alter the effect of the repair. Paired differences are used throughout because each seed contributes a raw, box-only, and fully repaired field from the same trained network, which removes between-seed architecture variability from the step-wise comparison.

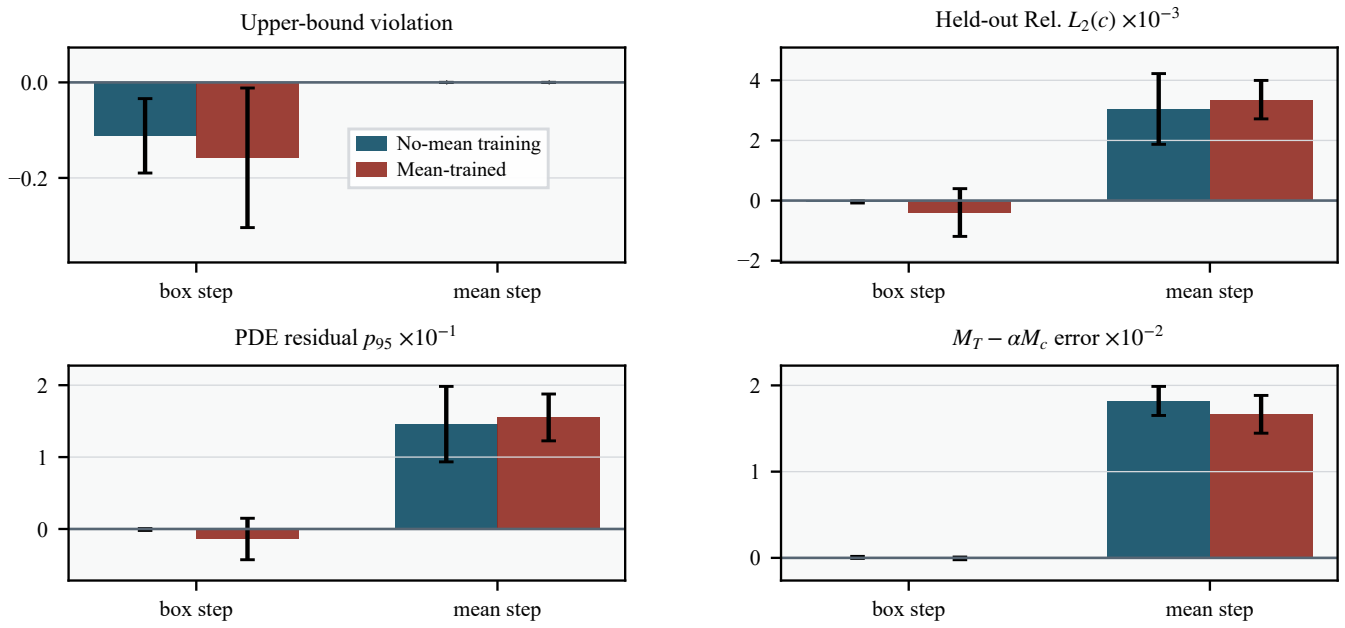


Fig. 2. Paired effect of each repair step over ten seeds. Bars show mean difference and whiskers show 95% CI. For the box step only the bound-violation effect is resolvable from zero; the increases in the other three metrics occur in the mean step.

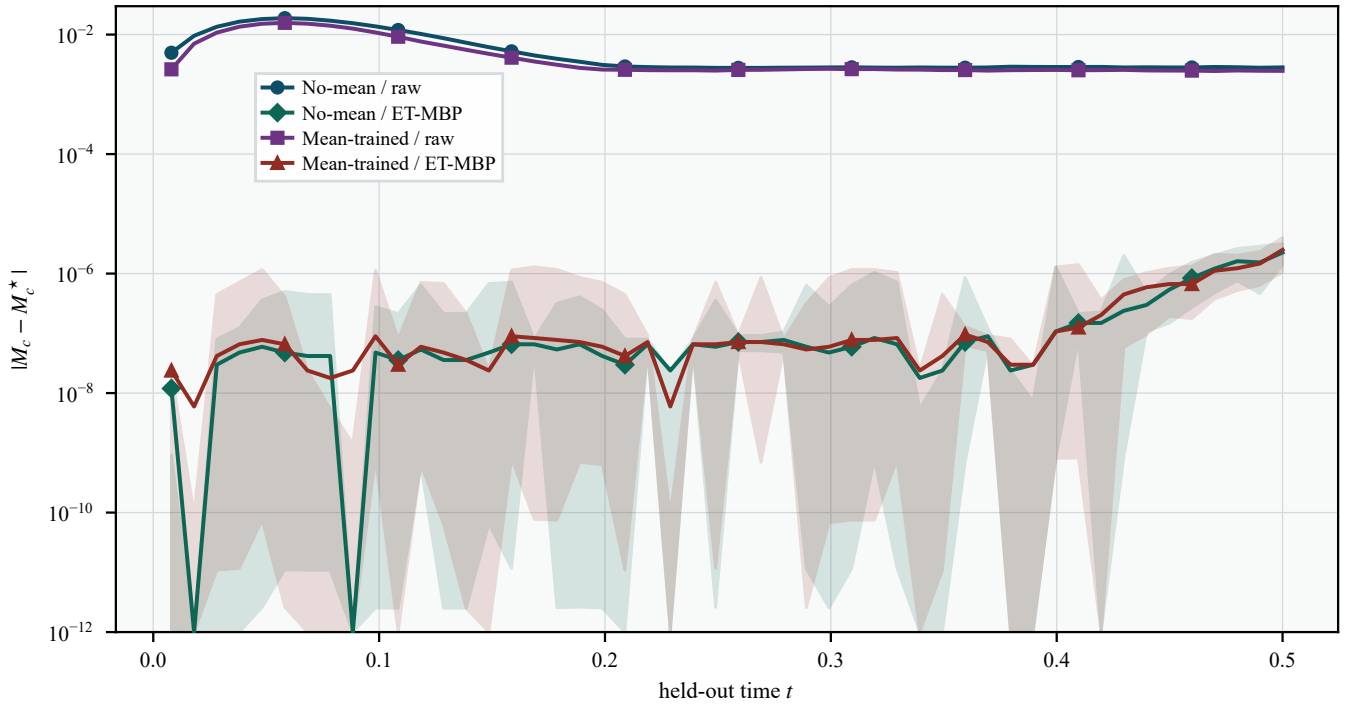


Fig. 3. Closure residual on the 50 time slices excluded from labeled training. Curves show the ten-seed mean; light bands show ±1 standard deviation on ET-MBP curves only. Box-only curves coincide with the raw curves and are omitted.

Figure 3 shows that the low post-repair residual is algebraic and persists on unseen labeled times. The raw curves stay much closer to the FD mean because their closure drift partly offsets the first-moment target error, as Eq. (9) permits.

Numerically converged capped-simplex projection drives the closure residual below 10^{-12}, yet its held-out Rel.$L_2(c)$, E_{true}, PDE p_{95}, and invariant error agree with the 60-step results to the displayed precision. The cost of the mean step is set by the

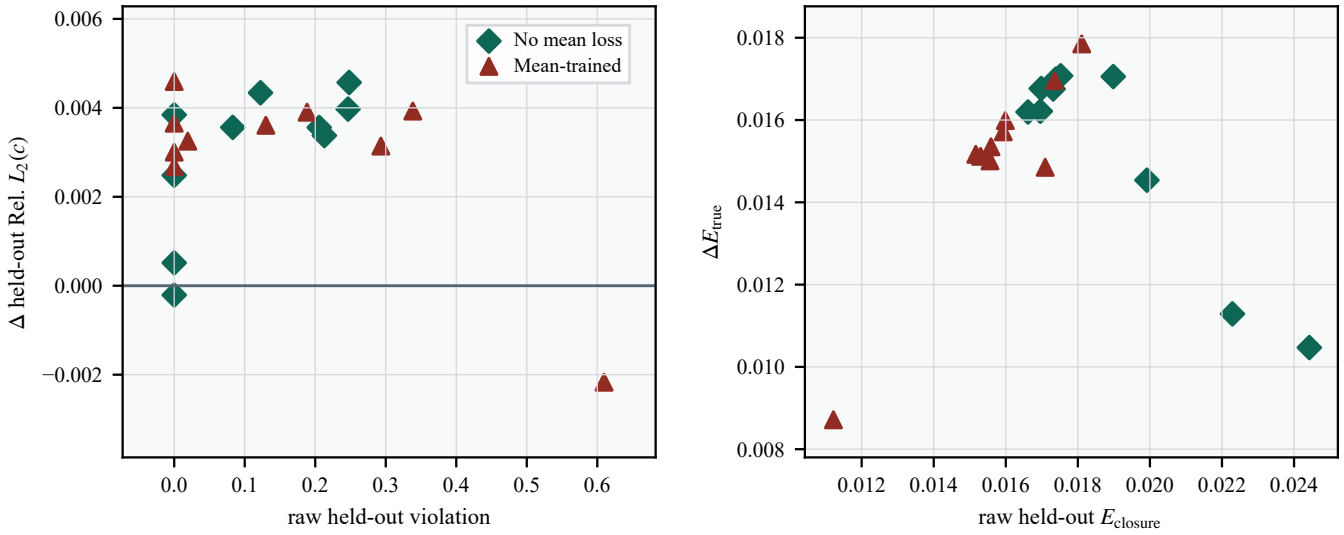


Fig. 4. Per-seed effect of the full repair. Left: raw held-out violation versus $\Delta\text{Rel.}L_2(c)$; right: raw held-out E_{closure} versus ΔE_{true}. A point at zero means the metric was left untouched; a positive value means the error grew.

prescribed target rather than by incomplete alternating convergence.

Seed-level behavior accounts for the large variance of the raw violation. Some saved networks already lie inside the box and gain no feasibility, whereas others carry a substantial upper-bound excess; see the left panel of Fig. 4. The right panel shows that the increase in E_{true} is nevertheless positive for every seed and both training groups, since it reflects the target rather than the network. A deployment trigger is better keyed to the measured bound violation and to the reliability of the target than to the nominal stiffness parameter. Neither step is expensive: on one CPU, repairing all 250 slices takes 0.19 ms for the 60-step operator and 0.00 ms for the box step alone.

IV. Discussion

Splitting the box and mean steps clarifies mean–box post-processing more than a single before/after comparison. On this benchmark the box step removed every held-out bound violation with no resolvable penalty in accuracy, PDE residual, or the coupled-mean invariant. The mean step carried the measurable cost of the composite repair, including a PDE residual increase not explained by the algebraic identities in Section II-B, because it matched a first-moment closure whose bias E_{model} (4.21×10^{-3}) already exceeded the raw held-out E_{true} (4.03×10^{-3}).

In deployment, box clipping suits bound violations on an exported field; mean adjustment is warranted only when the prescribed spatial mean is trustworthy. Joint T–c repair and variance-aware targets are natural extensions and remain open directions. The benchmark uses one stiff parameter set, a single finite-difference reference, and an interleaved temporal holdout; the box-step finding also reflects reference saturation at the upper bound.

V. Conclusion

We separated box clipping from mean adjustment in post-training repair for a stiff periodic heat–species reaction–diffusion PINN and audited the split on held-out time slices with ten paired seeds. Box clipping alone restored bound feasibility without measurable cost in the accuracy, PDE, and invariant metrics tracked here, whereas every measurable cost arose from mean adjustment and tracked the prescribed closure target rather than incomplete alternating convergence. Reporting the two steps separately gives a clearer breakdown of post-training repair than a single before/after comparison.

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